

1.1.13 Karnaugh Maps (K-Maps)

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

CS XI

Chapter 1: Computer Systems

Lecture 12 of 20

★ Golden

1.113 Karnaugh Maps (K-Maps) ★ Golden

LEARN IT

A Karnaugh map (K-map) is a graphical tool used to **simplify Boolean expressions**. It is a grid of cells — one cell for each row of the truth table. The cells are arranged so that **neighbouring (adjacent) cells differ by only ONE bit**. This ordering is called **Gray code**. You write the output (1 or 0) of each row into its cell, then **group the 1s** in rectangles of size 1, 2, 4 or 8 (powers of 2). Each group becomes one simple term. This simplifies the equation without long algebra.

Two-variable K-map (A, B)

2 variables need $2^2 = 4$ cells.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Three-variable K-map (A, B, C)

3 variables need $2^3 = 8$ cells. The columns follow the Gray code sequence **00, 01, 11, 10** (not 00, 01, 10, 11). This makes adjacent cells differ by only one variable.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	m_0 (A'B'C')	m_1 (A'B'C)	m_2 (A'BC)	m_3 (A'BC')
A = 1	m_4 (AB'C')	m_5 (AB'C)	m_6 (ABC)	m_7 (ABC')

LEARN IT

Grouping rules

1. Group only 1s. Groups must be rectangles of size 1, 2, 4 or 8 — never 3, 5 or 6.
2. Make groups **as large as possible**. Bigger groups give simpler terms.
3. Use **as few groups as possible**, but every 1 must be in at least one group.
4. **Overlapping is allowed** — a 1 may be used in more than one group.
5. **Wrapping around the edges is allowed**: a cell in the far-left column is adjacent to the cell in the far-right column (BC = 00 is next to BC = 10).
6. To read a group: variables that **change** (0 to 1) inside the group are **eliminated**. Variables that stay the same are kept — plain if they stay 1, complemented if they stay 0.

Group size tells you how many variables disappear: in a 3-variable map, a group of 2 leaves 2 variables, a group of 4 leaves 1 variable, and a group of 8 means $Y = 1$.

WORKED EXAMPLE

K-map example 1 (2 variables): simplify $Y = A'B + AB$.

Step 1: The minterms are A'B (cell 1) and AB (cell 3). Put 1 in those cells, 0 elsewhere.

	B = 0	B = 1
A = 0	0	1
A = 1	0	1

Step 2: The two 1s are in the same column, one above the other. Make one group of 2.

Step 3: Inside the group, A changes (0 then 1) → eliminate A. B stays 1 → keep B.

Result: $Y = B$. This agrees with the algebra in 11.4 and the minterm table in 11.8.

WORKED EXAMPLE

K-map example 2 (2 variables, overlapping): simplify $Y = A'B' + A'B + AB$.

Step 1: Put 1s in cells 0, 1 and 3.

	B = 0	B = 1
A = 0	1	1
A = 1	0	1

Step 2: Group 1 = the top row (cells 0 and 1). A stays 0, B changes → term A' .

Step 3: Group 2 = the right column (cells 1 and 3). A changes, B stays 1 → term B. Cell 1 is used twice — overlapping is allowed.

Result: $Y = A' + B$. Check: $A' + B$ is 0 only when $A = 1$ and $B = 0$, which is exactly the one 0 cell (cell 2). ✓

WORKED EXAMPLE

K-map example 3 (3 variables): simplify the majority function $Y = \sum m(0, 2, 5, 6, 7) = A'BC + AB'C + ABC' + ABC$.

Step 1: Fill the map. Careful — the columns are in Gray order 00, 01, 11, 10, so m_5 is in the third column and m_6 in the fourth.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	0	1 (m_2)	0
A = 1	0	1 (m_5)	1 (m_6)	1 (m_7)

Step 2: No group of 4 is possible (there is no 2×2 block or full row of 1s). So make groups of 2, each using m_i :

• Group 1: $m_5 + m_7$ (column BC = 11). A changes; B = 1, C = 1 stay → **BC**.

• Group 2: $m_6 + m_7$ (row A = 1, columns 01 and 11). B changes; A = 1, C = 1 stay → **AC**.

• Group 3: $m_2 + m_6$ (row A = 1, columns 11 and 10). C changes; A = 1, B = 1 stay → **AB**.

Result: $Y = AB + BC + AC$. Four 3-variable terms (needing 4 three-input AND gates) became three 2-variable terms.

Check one row: A=1, B=0, C=1 → AB = 0, BC = 0, AC = 1 → $Y = 1$. In the truth table, row 5 is 1. ✓

WORKED EXAMPLE

K-map example 4 (3 variables, wrap-around): simplify $Y = \sum m(0, 2, 4, 6)$

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	1	0	0	1
A = 1	1	0	0	1

Step 1: The 1s are in the far-left column (BC = 00) and the far-right column (BC = 10).

Step 2: Because the map wraps around, these two columns are adjacent. Make **one group of 4**: m_0, m_4, m_2, m_6 .

Step 3: In this group A changes (0, 1) and B changes (0, 1). Only C stays the same: C = 0 → term C' .

Result: $Y = C'$. Without wrapping you would get two groups (B'C' + BC'), which is correct but not the simplest.

WORKED EXAMPLE

K-map example 5 (3 variables): simplify $Y = \sum m(1, 4, 5, 6, 7)$ — this is the output of $Y = A + B'C$ from 11.8.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	0	0
A = 1	1	1	1	1

Step 1: Group of 4: the whole row A = 1 (m_4, m_5, m_6, m_7). B and C both change; A stays 1 → **A**.

Step 2: m_1 is still alone. Pair it with m_5 below it (column BC = 01). A changes; B = 0, C = 1 stay → **B'C**.

Result: $Y = A + B'C$ — the K-map confirms the original expression was already in its simplest form.

WORKED EXAMPLE

Going further — a 4-variable K-map: simplify $Y = \sum m(0, 2, 4, 5, 7, 8, 10, 12, 14)$. (Not required by the slides, but a good test of the same rules.) 4 variables need 16 cells; both rows (AB) and columns (CD) use Gray order 00, 01, 11, 10.

	CD = 00	CD = 01	CD = 11	CD = 10
AB = 00	1 (m_0)	0	0	1 (m_{12})
AB = 01	0	1 (m_5)	1 (m_7)	0
AB = 11	0	1 (m_{13})	1 (m_{14})	0
AB = 10	1 (m_8)	0	0	1 (m_{10})

Step 1: The centre 2×2 block (m_5, m_7, m_{13}, m_{14}) is a group of 4. A and C change; B = 1 and D = 1 stay → **BD**.

Step 2: The four corners (m_0, m_8, m_{12}, m_{10}) are adjacent through wrap-around (left-right and top-bottom). A and C change; B = 0 and D = 0 stay → **B'D'**.

Result: $Y = BD + B'D'$ (this is B XNOR D). Eight 4-variable minterms became two 2-variable terms.

REMEMBER IT

Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges touch. For reading groups: **"If it changes, it goes."**

COMMON MISTAKE

Most common K-map mistakes: (1) writing the columns as 00, 01, 10, 11 — this puts m_6 and m_4 in the wrong places; (2) making groups of 3 or 6 — only 1, 2, 4, 8 are allowed; (3) grouping diagonally — groups must be rectangles of side-by-side cells; (4) stopping at small groups when a bigger group (or a wrap-around group) is possible.

آرڈو میں سمجھیں

کارنو مپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔ اس کی ساتھ والے خانوں میں صرف ایک بیت کا فرق ہوتا ہے (گریہ کوڈ: 00, 01, 11, 10)۔ والے خانوں کو 0, 1, 2 یا 4 کے گروپوں میں جتنا بڑا ممکن ہو جوڑیں؛ گروپ ایک دوسرے پر چڑھ سکتے ہیں اور کناروں سے گھوم کر بھی بن سکتے ہیں۔ گروپ کی اندر جو منطقی بدلتا ہے وہ ختم ہو جاتا ہے۔

EXAM TIP

K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s too), circle the groups clearly, write the term for each group beside it, and give the final simplified expression. Examiners often add "draw the logic diagram of the result" — so practise drawing it too.

CHECK YOURSELF

1. In a 3-variable K-map, which column is next to BC = 01 on the right?

2. Simplify $Y = \sum m(2, 3)$ for two variables A, B.

3. Simplify $Y = \sum m(4, 5, 6, 7)$ for three variables.

Answers: 1. BC = 11. 2. Cells 2 and 3 are the row A = 1; B changes → $Y = A$. 3. The whole row A = 1 is a group of 4 → $Y = A$.

Next lecture: LogiSim Evolution: Simulation Tool